

CS445 Final Project Report

1. Introduction

The objective of our project was stance detection. In this project, stance detection was approached in two stages: Target Classification and Stance Detection. In the first stage, the model predicted the topic or "target" of the tweet, such as Atheism or Hillary Clinton. In the second stage, using the predicted target, the second model determined the stance of the tweet towards the target. In the final result, both Target and Stance were estimated. We found that logistic regression was the best model for target classification while finetuned BERT was the best model for stance detection. We replicated the paper from the SemEval-2016 task and achieved a greater f1-score, macro precision, and macro recall with our finetuned BERT model.

2. Methodology

We have chosen to implement the model with the highest F-score in SemEval-2016 Task 6, which was Mohammed et al.'s 2017 paper "Stance and Sentiment in Tweet", in an effort to reproduce its results. Then we attempted to improve their performance by using BERT embeddings and other classifiers.

Dataset Preprocessing

The project used the SemEval-2016 Stance Dataset, which was designed for stance detection tasks. It consists of tweets labeled with targets and their corresponding stances.

Below, the first five rows of the dataset and their columns are displayed:

Tweet	Target	Train/Test	Stance	Opinion Tow	Sentiment labels
If abortion is not wrong, then nothing is wrong. Powerful words from Blessed Mother Teresa	Legalization	Train	AGAINST	Target	pos
Mary, Help of Christians persecuted everywhere, pray for us! #HolyLove #UnitedHearts #M	Legalization	Train	AGAINST	Other	pos
TY Michael @ASavageNation "I'd do anything to help him; he's right, he's telling the truth" @Donald Trum		Test	FAVOR	Target	pos
1 Cor 15:58 ...stand firm...Always give yourselves fully to the work of the #Lord...your labor	Atheism	Train	AGAINST	Other	pos
1 Corinthians 13:13 "So now faith, hope, and love abide, these three; but the greatest of the	Atheism	Train	AGAINST	Target	pos

The columns include Tweet, Target, Train/Test, Stance, Opinion Toward, and Sentiment labels. The tweet section contains text of the tweet. The target section specifies what the topic is about.

The target, "Donald Trump," was excluded from the dataset because it was not in the training data. Afterward, in the preprocessing section, the tweets were converted to lowercase letters, words starting with @ were removed from the rows, special characters were cleaned, and only characters such as letters, numbers, spaces, '-', '_', and '#' was allowed, and extra spaces were cleaned. Finally, the Preprocessed_Tweet column was created with changes. Target and Stance columns were converted to numeric values to be included in the models.

Training, Validation and Test Splits

The training data was split into 64% training, 16% validation, and 20% test sets for both target and stance classification tasks.

Tokenize and Sequence Texts

The BERT tokenizer transforms textual data into numerical sequences for the model to process effectively. Each word is converted to an integer index based on its frequency. Sequences are padded or truncated to a fixed length. This ensures uniform input dimensions and preserves semantic structure while optimizing.

Models

We tried multiple models with our data besides replicating our chosen paper.

Recurrent Neural Network (Serhan):

Target Classification (RNN)

- **Embedding Layer:** Vocabulary size of 10,000 and embedding dimension of 128
- **LSTM Layer:** 128 units
- **Dense Layers:** A ReLU-activated layer with 128 units and a softmax output layer
- **Loss:** Sparse categorical cross entropy
- **Optimizer:** Adam optimizer with 0.001 learning rate
- **Epochs:** 20 epochs
- **Batch Size:** 32

Stance Classification (RNN)

- **Embedding Layer:** Vocabulary size of 10,000 and embedding dimension of 128
- **LSTM Layer:** 128 units
- **Concatenation:** Combining the LSTM output with the target one-hot vector
- **Dense Layers:** ReLU-activated hidden layer with 64 units and a softmax output layer
- **Loss:** Sparse categorical cross entropy
- **Optimizer:** Adam optimizer with a learning rate of 0.008
- **Epochs:** 8
- **Batch Size:** 32

SVM with Word2Vec and BERT Embeddings (Bengi)

I've replicated Mohammad et al.'s model, which consisted of a linear SVM for stance classification with 1,2,3 word ngrams; 2, 3, 4, 5 character ngrams; presence of target and Word2Vec embeddings. Then I used a linear SVM classifier again with BERT embeddings instead of Word2Vec embeddings. I also used the same features with SGD and Random Forest Classifier with 200 estimators to test the effectiveness of these features on other classifiers. I also finetuned a BERT model to test if BERT embeddings performed better with its own model, and when it did perform better for 10 epochs with 8 batch size.

Feedforward Neural Network (Zeynep)

Target Classification

- **Architecture:** An input layer for BERT embeddings, a hidden layer with 128 ReLU-activated units, and an output layer with softmax activation for multi-class classification
- **Regularization:** A dropout layer with a 30% rate was used for regularization
- **Loss Function:** Cross-entropy loss for multi-class classification.
- **Optimizer:** Adam optimizer with a learning rate of 0.0001
- **Performance:** Achieved a macro F1-score of 0.69 on the test set for target classification.
- **Epochs:** 10
- **Batch Size:** 32

Stance Classification

- **Features:** Text features combined with the target context via BERT embeddings (bert-base-uncased, CLS token)
- **Loss Function:** Cross-entropy loss for multi-class classification.
- **Performance:** Achieved a weighted F1-score of 0.64 on the test set for stance classification.
- **Epochs:** 10
- **Batch Size:** 32

Naive Bayes with BERT Word Embeddings (Nida)

Target Classification

- **Features:** TF-IDF vectorization with n-grams (1,2) and a vocabulary size of 10,000.
- **Model:** Logistic Regression with hyperparameter tuning.
- **Optimization:** GridSearchCV was used to optimize parameters.
- **Performance:** Achieved an accuracy of **72%** for target classification.

Stance Classification

- **Features:** BERT embeddings (bert-base-uncased, CLS token, embedding dimension 768).
- **Model:** Multinomial Naive Bayes
- **Normalization:** MinMaxScaler was applied to BERT embeddings for consistency across models.
- **Performance:** Achieved an **overall stance detection F1-Score of 54%**.

3. Results

Model	Test F1-Score (Target)	Test Macro Precision (Target)	Test Macro Recall (Target)	Test F1-Score (Stance)	Test Macro Precision (Stance)	Test Macro Recall (Stance)
Recurrent Neural Network	0.63	0.65	0.64	0.55	0.49	0.50
Feedforward Neural Network	0.69	0.65	0.64	0.64	0.65	0.64
Logistic Regression (Target Classification) + Naive Bayes (Stance Detection)	0.70	0.71	0.70	0.54	0.55	0.55
Paper Reproduced (SVM + Word2Vec + ngrams + Target)	-	-	-	0.59	0.58	0.60
SVM + BERT + ngrams + Target	-	-	-	0.61	0.60	0.62
Finetuned BERT	-	-	-	0.64	0.64	0.66

We first reproduced the work of Mohammed et al. and achieved 0.59 F1-score, 0.58 macro precision, and 0.60 macro recall.

Finetuned Bert had the best performance for stance classification. It had 0.64 f1-score, 0.64 macro precision, and 0.66 macro recall. Its confusion matrix is included in the appendix. It was better at classification compared to Naive Bayes and the paper we reproduced. Its f1-score, macro precision, and macro recall were all better. The precision-recall curves of each stance category of BERT is also included in the appendix.

We also tried different models with BERT embeddings and their statistics are also included in the appendix. We also found that models performed differently for each target class, and models that were trained on only a single target class often performed better than a model that was trained on all data despite the lesser amount of data. These statistics are also included in the appendix.

We also observed that models that only used BERT embeddings like Naive Bayes and Recurrent Neural Networks performed slightly worse than the Finetuned BERT model, though they mostly performed better than the paper reproduction, which was one of our objectives. The Feedforward Neural Network also performed nearly as well as the finetuned BERT model, and its macro precision was slightly better while BERT's recall was better.

Logistic Regression was the best model for target classification with 0.70 f1-score and recall and 0.71 precision.

4. Discussion

Dataset Selection

“Donald Trump” was missing from the training data, so we removed it from the test data as the target classification would not work for unseen targets. The stance for each target category was also not equal, which made predicting each stance equally well more difficult.

Methodology Choices

- **Advantages:**
 - Use of LSTM layers to capture sequential dependencies.
 - The functional API model allowed the integration of additional features (target one-hot).
 - Random Forest Classifier and Naive Bayes Classifiers were the fastest to train and predict compared to other models we tried.
- **Disadvantages:**
 - Model complexity increased training time and resource requirements. For example, training SVMs took longer than training
 - Computing BERT embeddings was slower compared to Word2Vec embeddings
 - Finetuning BERT would have been very time-consuming without utilizing Google Colab’s GPUs.

Result Comparison

- The proposed models moderately improved over the baseline but fell short of state-of-the-art benchmarks, likely due to limited data size and model capacity.
- Our BERT model and some of our models were better than the reproduced paper.

Limitations

- Imbalanced classes impacted the performance of some stances.
- Lack of hyperparameter tuning constrained optimization.

Improvements

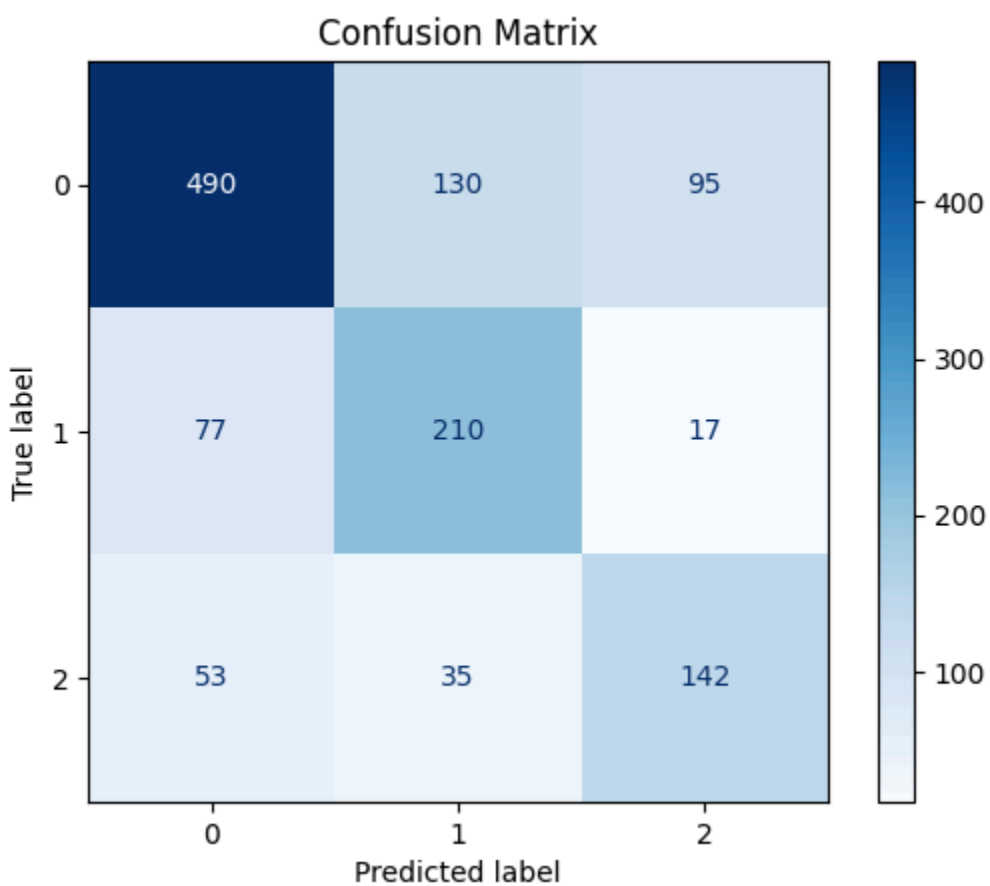
- Larger datasets could mitigate class imbalance.
- Hyperparameter optimization and more advanced architectures (e.g., transformers) could have enhanced our performance.

5. Conclusion

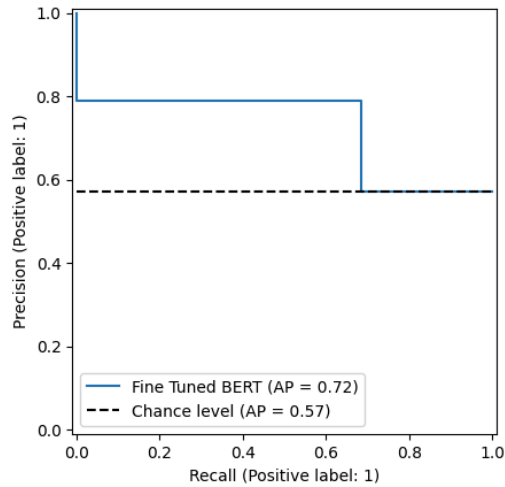
In short, we observed that logistic regression was the best model for stance target classification while BERT was the best for stance classification. By finetuning BERT on the training data, we achieved a slightly better performance than the model which replicated the work of Mohammed et al.’s 2017 paper “Stance and Sentiment in Tweet”.

6. Appendix

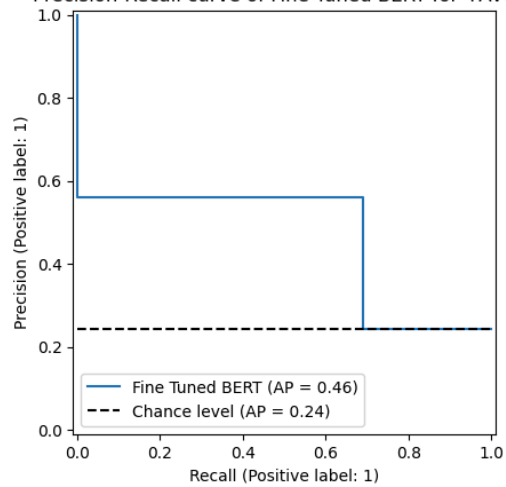
Confusion Matrix of Fine Tuned BERT where 0 is “AGAINST”, 1 is “FAVOR”, and 2 is “NONE”.



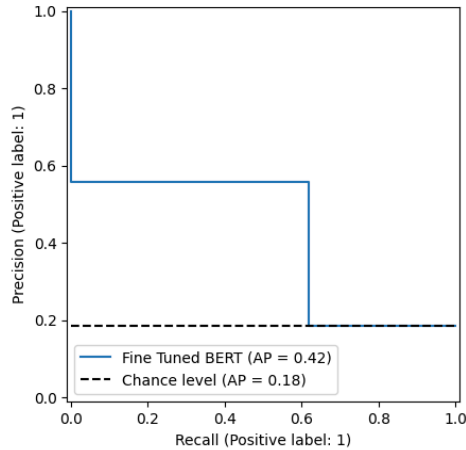
Precision-Recall curve of Fine Tuned BERT for 'AGAINST'



Precision-Recall curve of Fine Tuned BERT for 'FAVOR'



Precision-Recall curve of Fine Tuned BERT for 'NONE'



Accuracies, Micro F1 and Macro F1 for only 'Favor' and 'Against', Macro Precision and Macro Recall of SVM, SGD, Random Forest and BERT

Model	Accuracy	Micro F1	Macro F1	Macro Precision	Macro Recall
Paper's Results (Mohammed et a.l)		0.703	0.590		
Paper Reproduced (SVM + Word2Vec + ngrams + Target)	0.63	0.66	0.64	0.58	0.60
SVM + BERT + ngrams + Target	0.64	0.67	0.65	0.60	0.62
SGD + BERT + ngrams + Target	0.64	0.66	0.63	0.60	0.63

Random Forest + BERT + ngrams + Target	0.67	0.71	0.61	0.70	0.53
Finetuned BERT	0.67	0.67	0.64	0.64	0.66

Micro F1 scores of only Favor and Against

Model	Atheism	Climate Change	Feminist Movement	Hillary Clinton	Abortion	Average Micro F1
Paper Reproduced (SVM + Word2Vec + ngrams + Target)	0.71	0.72	0.59	0.66	0.68	0.67
SVM + BERT + ngrams + Target	0.73	0.76	0.59	0.67	0.67	0.68
SGD + BERT + ngrams + Target	0.73	0.74	0.57	0.66	0.66	0.67
Random Forest + BERT + ngrams + Target	0.77	0.66	0.71	0.67	0.74	0.71

Micro F1 scores of only Favor and Against of Models that were only trained on one class's data

Model	Atheism	Climate Change	Feminist Movement	Hillary Clinton	Abortion	Average Micro F1
Paper Reproduced (SVM + Word2Vec + ngrams + Target)	0.77	0.83	0.60	0.70	0.70	0.72
SVM + BERT + ngrams + Target	0.80	0.84	0.62	0.72	0.69	0.734
SGD + Word2Vec + ngrams + Target	0.77	0.82	0.56	0.72	0.69	0.712
SGD + BERT + ngrams + Target	0.78	0.83	0.59	0.71	0.65	0.712
Random Forest + Word2Vec + ngrams + Target	0.75	0.79	0.59	0.69	0.69	0.702
Random Forest + BERT + ngrams + Target	0.77	0.85	0.68	0.69	0.74	0.746

7. References

- Bøhler, H., Asla, P. F., Marsi, E., & Sætre, R. (2016). IDI@NTNU at SemEval-2016 Task 6: Detecting Stance in Tweets Using Shallow Features and GloVe Vectors for Word Representation. *Proceedings of the International Workshop on Semantic Evaluation (SemEval-2016)*, San Diego, California. <https://doi.org/10.18653/v1/S16-1072>
- Mohammad, S., Kiritchenko S., Sobhani P & Kiritchenko, S. (2017). Stance and Sentiment in Tweets. *ACM Trans. Internet Technol.* 17, 3, Article 26 (August 2017), 23 pages. <https://doi.org/10.1145/3003433>.